

Cofinal compact-window Weil positivity

A self-contained finite-matrix problem

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Abstract

This note offers a precisely localized finite-matrix *form of* the Riemann hypothesis, never a lower-strength statement. It asks whether an explicit bordered dual resolvent is at least one half of the identity for infinitely many indices. The matrix is built from the von Mangoldt comb, the archimedean Weil kernel, one fixed smooth comparison comb, and the reserve 5/7. Round 469 binds the strength reading: because the compact-window floor $\lambda_*(L)$ is nonincreasing, positivity on any unbounded cofinal sequence is the full Weil criterion after the two internal identifications and the three separately named external bridges. No such proof is claimed. What is supplied is an exact finite formulation, a certified finite census, the r467/r468 anatomy of its correlation wall, and a scramble filter that rules out whole classes of attacks. A distinct fixed- L prize is stated separately; unlike the cofinal problem it is not RH-equivalent.

Grades. [E] means an exact finite identity or a proved elementary statement; [C] means a sealed finite census; [O] means open. No census is extrapolated and no open statement is promoted.

Strength warning (binding r469 reading). The target below is not a tractable fragment offered beneath RH. Modulo the explicitly listed internal and external bridges, the monotonicity of $\lambda_*(L)$ makes “good on an unbounded subsequence” equivalent to positivity at every finite support length, hence to the full Weil criterion. The document locates that problem in one explicit family of finite matrices; it does not prove it and does not count its finite certificates as progress toward RH. Source: r469, commit f455a22a.

1 The named sequence

For $k \geq 1$ set

$$a_k = 2^k, \quad r_k = \lfloor \sqrt{k} \rfloor, \quad m_k = k2^{r_k} - 1, \quad M_k = m_k + 1, \quad L_k^{\text{grid}} = 2M_k - 2, \quad (1)$$

and

$$\Delta_k = \frac{\log a_k}{M_k} = 2^{-r_k} \log 2, \quad N_k = \frac{M_k}{2} = k2^{r_k-1}. \quad (2)$$

Here M_k is even for the range of interest. The folded grid has $m_k = L_k^{\text{grid}}/2$ points

$$\theta_{k,j} = \frac{2\pi j}{L_k^{\text{grid}}}, \quad x_{k,j} = \cos \theta_{k,j}, \quad 1 \leq j \leq m_k.$$

The last point is its own mirror and receives half the generic fold weight.

2 The faithful fold

Write

$$T_\Delta(t) = \max\{0, 1 - |t|/\Delta\}$$

and let $\Lambda(n) = \log p$ if $n = p^\ell$ is a prime power, and 0 otherwise. The prime lags are the finite sums

$$c_{k,i}^P = - \sum_{2 \leq n \leq a_k^2} \frac{\Lambda(n)}{\sqrt{n}} [T_{\Delta_k}(i\Delta_k - \log n) + T_{\Delta_k}(i\Delta_k + \log n)], \quad 0 \leq i < M_k. \quad (3)$$

The reflected term is nonzero only when $\log n < \Delta_k$.

For completeness, define the archimedean lag kernel without a numerical quadrature convention. Put γ_E for Euler's constant. For $s \geq \Delta > 0$ let

$$A_\Delta(s) = - \int_{s-\Delta}^{s+\Delta} T_\Delta(s-w) \frac{e^{-w/2}}{1 - e^{-2w}} dw. \quad (4)$$

For $0 \leq s < \Delta$, put $t = T_\Delta(s)$, $S_s(w) = \frac{1}{2}(T_\Delta(s-w) + T_\Delta(s+w))$, and define

$$\begin{aligned} A_\Delta(s) = & -(\gamma_E + \log \pi)t + 2 \int_0^{s+\Delta} \frac{te^{-2w} - S_s(w)e^{-w/2}}{1 - e^{-2w}} dw \\ & - t \log(1 - e^{-2(s+\Delta)}). \end{aligned} \quad (5)$$

The apparent singularity at $w = 0$ is removable. Set $c_{k,i}^A = A_{\Delta_k}(i\Delta_k)$ and $c_{k,i} = c_{k,i}^A + c_{k,i}^P$.

The circulant spectral density and signed folded weights are

$$d_{k,j} = c_{k,0} + 2 \sum_{i=1}^{M_k-2} c_{k,i} \cos(i\theta_{k,j}) + c_{k,M_k-1} \cos((M_k-1)\theta_{k,j}), \quad (6)$$

$$w_{k,j} = \frac{2}{L_k^{\text{grid}}} (1 - \cos \theta_{k,j}) d_{k,j} \times \begin{cases} 1, & j < m_k, \\ \frac{1}{2}, & j = m_k. \end{cases} \quad (7)$$

Define positive measures on $[-1, 1]$ by the sign split

$$\mu_k = \sum_j \max(w_{k,j}, 0) \delta_{x_{k,j}}, \quad \nu_k = \sum_j \max(-w_{k,j}, 0) \delta_{x_{k,j}}. \quad (8)$$

Equations (1)–(8) are the faithful production order: tents, total lags, circulant density, fold, sign split. No atom-table cutoff occurs; all prime powers through a_k^2 are included.

3 The border, budget, and $R^\dagger$

The border is fixed by the following smooth comparison comb. Let $\eta = 10^{-2}$,

$$u_\ell = \ell\eta, \quad m_\ell = 2e^{u_\ell/2}\eta, \quad 0 < u_\ell < 2 \log a_k.$$

Replace the prime-power positions and masses in (3) by (u_ℓ, m_ℓ) , equivalently use

$$c_{k,i}^\sigma = -\frac{1}{2} \sum_\ell m_\ell [T_{\Delta_k}(i\Delta_k - u_\ell) + T_{\Delta_k}(i\Delta_k + u_\ell)],$$

add the same $c_{k,i}^A$, and apply (6)–(7). Retaining the signs gives a finite signed measure σ_k .

Assume the moment Gram of μ_k is positive through degree $N_k - 1$, and let $p_{k,0}, \dots, p_{k,N_k-1}$ be its real orthonormal polynomials. Define

$$b_{k,i} = \int p_{k,i}(x) d\sigma_k(x), \quad B_k^{\text{bud}} = \sum_{i=0}^{N_k-2} b_{k,i}^2 + \frac{5}{7}, \quad \beta_k = \frac{b_k}{\sqrt{B_k^{\text{bud}}}}. \quad (9)$$

Thus both numerical conventions in the border are explicit: $\eta = 0.01$ and the reserve is exactly $5/7$.

Here is an intrinsic definition of the dual block. Let $X_k = \text{supp}(\mu_k + \nu_k)$, $P_{X_k}(x) = \prod_{\xi \in X_k} (x - \xi)$, and, for $\xi \in X_k$, let

$$u_{k,\xi}^\vee = \frac{1}{(\mu_k + \nu_k)(\{\xi\}) P'_{X_k}(\xi)^2}.$$

Let $\Pi_{k,N_k-1}^\vee$ be the dressed Christoffel–Darboux projection of rank $N_k - 1$ for this reciprocal measure, and let R_k be its restriction to $Y_k = \text{supp } \nu_k$. For $y \in Y_k$, set

$$(v_k)_y = \sqrt{\nu_k(\{y\})} \sum_{i=0}^{N_k-1} (\beta_k)_i p_{k,i}(y), \quad (D_k)_{yy} = \text{sgn } P'_{X_k}(y), \quad \tilde{v}_k = D_k v_k,$$

and $\gamma_k = \|\beta_k\|^2$. The bordered dual resolvent is

$$R^\dagger(W_k) = \begin{pmatrix} R_k^{-1} & \tilde{v}_k \\ \tilde{v}_k^T & 1 + \gamma_k \end{pmatrix}^{-1}. \quad (10)$$

The definition applies whenever the displayed Gram and inverse conditions hold. Their failure at infinitely many indices would itself refute the requested statement.

4 The problem in three coordinates

Problem 1 (Finite-matrix form). [O] *Prove*

$$R^\dagger(W_k) \succeq \frac{1}{2}I \quad \text{for infinitely many } k \text{ in the sequence (1).}$$

This semidefinite statement, rather than a strict asymptotic margin, is the requested finite-matrix form. It has two useful equivalent coordinates.

Classical coordinate. For a compactly supported $h \in L^2(\mathbb{R})$ put $g = h * \tilde{h}$ and $Q_W(h) = W(g)$, the additive-log Weil quadratic form. Define

$$\lambda_*(L) = \inf_{\substack{0 \neq h \in L^2(\mathbb{R}) \\ \text{supp } h \subset [-L, L]}} \frac{Q_W(h)}{\|h\|_2^2}.$$

The selected support scales are

$$U_k = N_k \Delta_k = \frac{k \log 2}{2}, \quad L_k = \frac{U_k}{2} = \frac{k \log 2}{4}. \quad (11)$$

After the internal fold/read and archimedean identifications, Problem 1 reads

$$\lambda_*(L_k) \geq 0 \quad \text{for an unbounded subsequence.}$$

Because the support spaces are nested, $\lambda_*(L)$ is nonincreasing; hence this is cofinal compact-window Weil positivity.

Spectral coordinate. Let

$$(\mathcal{B}_k)_{yi} = \sqrt{\nu_k(\{y\})} p_{k,i}(y), \quad C_k = \mathcal{B}_k^T \mathcal{B}_k, \quad u_k = \beta_k.$$

If $C_k e_{k,j} = \lambda_{k,j} e_{k,j}$ and $C_k \prec I$, then

$$q_k^\dagger := u_k^T (I - C_k)^{-1} u_k = \sum_j \frac{|\langle u_k, e_{k,j} \rangle|^2}{1 - \lambda_{k,j}}. \quad (12)$$

Finite duality and Schur complementation give

$$R^\dagger(W_k) \succeq \frac{1}{2}I \iff \begin{pmatrix} I - C_k & u_k \\ u_k^T & 1 \end{pmatrix} \succeq 0 \iff q_k^\dagger \leq 1 \quad (13)$$

under the stated nondegeneracy assumptions. Thus the sharp internal task is to bound the spectral measure of the explicit border vector near the saturating edge $\lambda = 1$.

5 What is known

5.1 Unconditional compact-support zones

For an autocorrelation $g = h * \tilde{h}$ one has $\hat{g} = |\hat{h}|^2 \geq 0$ [E]. Bombieri's classical small-support theorem gives strict Weil positivity when the length of the support interval of h is $U < \log 2$. A recent computational preprint reports the certified extension

$$Q_W(h) \geq 8.9 \cdot 10^{-18} \|h\|_2^2 \quad \text{for} \quad \text{supp } h \subset [-0.8, 0.8],$$

or autocorrelation support $U \leq 1.6$ [C]. This recent bound is not being treated as a classical theorem.

By (11), $U_5 = 1.732868 > 1.6$. Every measured selected window therefore lies beyond both unconditional zones. The crossing is part of the point: the finite race below is not explained by prime blindness or by the $U \leq 1.6$ certificate.

5.2 Finite selected-window census

k	m_k	N_k	Δ_k	U_k	race $q_k^\dagger$	raw norm bound	top overlap
5	19	10	.1732868	1.732868	.675060	4.825	$9.53 \cdot 10^{-3}$
6	23	12	.1732868	2.079442	.694907	14.227	$2.22 \cdot 10^{-6}$
7	27	14	.1732868	2.426015	.628736	47.207	$8.89 \cdot 10^{-4}$
8	31	16	.1732868	2.772589	.606566	93.728	$5.05 \cdot 10^{-4}$
9	71	36	.0866434	3.119162	.635369	724.586	$1.94 \cdot 10^{-5}$
10	79	40	.0866434	3.465736	.757783	1593.258	$4.15 \cdot 10^{-6}$
11	87	44	.0866434	3.812309	.715782	2755.648	$5.10 \cdot 10^{-6}$
12	95	48	.0866434	4.158883	.693564	4991.649	$1.54 \cdot 10^{-6}$

All eight races lie in $[0.606566, 0.757783]$ and are below 1 [C]. The spectral identity (12) agrees with the direct solve to $3.85 \cdot 10^{-14}$ and with the independently sealed race object to $1.89 \cdot 10^{-14}$ [C]. None of these numbers proves an infinite subsequence.

5.3 Exact identities available to an attack

Identity	One-line proof
$\Delta_k = 2^{-r_k} \log 2$ and $U_k = k \log 2/2$ [E]	Insert $a_k = 2^k$ and $M_k = k2^{r_k}$ into $\Delta_k = \log(a_k)/M_k$ and $U_k = N_k \Delta_k$.
$\#\{k : \lfloor \sqrt{k} \rfloor = r\} = 2r + 1$ [E]	The block is the integer interval $[r^2, (r+1)^2)$.
$\widehat{h * \widetilde{h}} = \widehat{h} ^2$ [E]	Use the convolution theorem and $\widehat{\widetilde{h}} = \overline{\widehat{h}}$.
Spectral race identity (12) [E]	Diagonalize the real symmetric matrix C_k .
Schur equivalence in (13) [E]	Take the Schur complement of $I - C_k$; finite particle-hole duality supplies the first equivalence.
$\lambda_*(L)$ is nonincreasing [E]	The admissible support subspaces are nested as L increases.
Two-band estimate [E]	For $P_\theta = \mathbf{1}_{(\theta,1)}(C_k)$, split (12) and bound the two denominators by $1 - \theta$ and $1 - \lambda_{\max}(C_k)$.

6 The separate fixed-window prize

This section has no cofinal quantifier. Interval arithmetic certifies the finite matrix inequality for every $k = 5, \dots, 16$ [E] (r465, commit 74bf4ef4; extension quoted in r468, commit 4a0cb4a0). The largest centred support half-width is

$$L_{16} = \frac{16 \log 2}{4} = 4 \log 2 = 2.772589\dots,$$

compared with 0.8 in the external computational certificate.

The following payoff is conditional, and its conditions are part of the same sentence: *if* `SelectedReadQuadraticRepresentation`, `arch_gauss_mellin_digamma_identity`, and fixed-support dyadic density/continuity are proved, *then* the finite certificates imply $\lambda_*(L_k) \geq 0$ for each certified $k \leq 16$. Closing those three individually finite or standard tasks would make this a new unconditional fixed-window theorem. It would remain strictly weaker than the cofinal problem and would not be RH-equivalent. Source and classification: r469, commit f455a22a.

7 The correlation wall

Measured anatomy. On $k = 5, \dots, 12$, the top MAIN eigenvector has 0.926–0.989 of its squared Fourier mass in the lowest quarter of frequencies, while the smooth ARCH control aligns with its top mode at 0.995813–0.998970. A deterministic source-position scramble raises the dangerous normalized top overlap by a factor $28.5\text{--}5.3 \cdot 10^5$. At fixed $k = 11$ geometry, the isolated $k = 10 \rightarrow 11$ prime octave moves $q^\dagger$ by $+0.2199$, while the complete step is -0.01180 . Thus the protective cancellation is not in either factor separately; it lies in the joint dependence of the border vector and the eigenbasis on the same arithmetic source [C]. Sources: r467, commit a0e9aa4d; r468, commit 4a0cb4a0.

Exact extremal coordinate. Put

$$\omega_k = \mu_k - \nu_k, \quad \mathcal{A}_k = \begin{pmatrix} I - C_k & u_k \\ u_k^T & 1 \end{pmatrix}.$$

Whenever the denominator is positive on $\mathcal{P}_{<N_k} \setminus \{0\}$,

$$q_k^\dagger = \frac{1}{B_k^{\text{bud}}} \sup_{0 \neq p \in \mathcal{P}_{<N_k}} \frac{(\int p d\sigma_k)^2}{\int p^2 d\omega_k}, \quad 1 - q_k^\dagger = \frac{\det \mathcal{A}_k}{\det(I - C_k)}. \quad (14)$$

The maximizer is the ω_k -Riesz representative of $p \mapsto \int p d\sigma_k$ [E]. Every failed pairing route replaces this *one* supremum by a product of two suprema—a data majorant times a mode majorant—and thereby discards the measured alignment. This is a diagnosis, not a bound. Source: r469, commit f455a22a.

Attacker’s pre-filter (methodological, not a theorem). Before attempting a proof, apply the deterministic scramble to its hypotheses. If the proposed bound is scramble-invariant, it must also absorb the observed factor $28.5\text{--}5.3 \cdot 10^5$ and is rejected before further work. A viable attack must identify a source-sensitive input that changes under this scramble and retains the joint dependence in (14). This filter does not assert that the scrambled value of $q^\dagger$ was measured; only the dangerous alignment was measured. Source: r469, commit f455a22a, based on r467, commit a0e9aa4d.

8 Compressed kill list

Rows 1–31 quote the sealed dossier aggregation r466 (commit 2b94dac8); row 32 is r467 (commit a0e9aa4d); row 33 is r468 (commit 4a0cb4a0). Each row records one killing pin, not a family theorem.

#	Route	Killing pin
1	L^∞/d_{arm} majorant	Taylor remainder / linear term ≥ 0.50
2	Fold-parity pairing	residual 0.512
3	Finite-head Chebyshev / uniform Z	χ mutants 1.021 and 0.892
4	Gershgorin/Schur assist	first useful depth only 0.11 N
5	OSC–Geronimus / Neumann R_2	quadratic/linear ratio 0.86
6	Checkerboard / M -matrix	sign agreement 0.504
7	Three-gap mask	8–18 distinct gaps
8	Isolation of fold mask	core pair count reaches 16
9	Source Weyl-energy decay	slope +0.542
10	Bulk FORM.T / FORM.P	repair rank 49, not 1
11	(P1) as construction class	$K/2 = 2$ fails on 9/42
12	Moments / Euler source Gram	residual 209
13	Ones mode / lift overflow	cos = 0.007
14	ν rank-one / Loewner sandwich	$N_\nu = 104$
15	Nyquist density 1/2	measured density 0.283
16	Literal $\Theta = C^T \Phi_N C$	relative mismatch 0.96
17	Unfitted Krein transport	residual 3.37
18	Test polynomial / Fourier / Bernstein	loss factor 90
19	Identity saturation $\ \mathfrak{T}_0\ = 1$	$C_{\min} - 1 = 4.57 \cdot 10^{-8}$
20	Hole top mode	leftover $\sigma = 1.024$
21	Top mode as edge defect	cosine 0.6766 versus 0.058
22	de Branges phase dominance	$yy_A = 3$
23	Uniform $\varphi_{bb} < 0$	vacuous overflow 6/14
24	Uniform denominator gap	reserve 0.105 $\rightarrow$ 0.038
25	Razor overlap as global coordinate	selected margin 0.617
26	$C_{\min}$ Nyquist face	cosine -0.007
27	Unsigned $ Z_{\text{loc}} $ triangle	loss 2.240–19.245
28	Chebyshev ψ / left–right Abel	pair count 1
29	Prime-blind prefix as mincut	world difference $3.33 \cdot 10^{-16}$
30	Raw resolvent norm	4.825 $\rightarrow$ 4991.649
31	Fixed- r blocks	exactly $2r + 1$ indices
32	Exact-coordinate Abel chain	factor 91.703 $\rightarrow$ 119482.632
33	Octave renormalization / block limit	+0.2199 isolated versus -0.01180 total

9 Numerical provenance

Every non-definitional decimal and campaign constant above is tied to a sealed source. Commit prefixes identify the immutable record used here.

Numbers	Sealed source
$\eta = 0.01$; reserve 5/7	r243 source, corpus commit 36211146; budget identity resealed r362, 62ca2bfa
$4 \cdot 10^5$, anchors 631/641, complete-comb repair	r458, b315ead4
$U < \log 2$; $U \leq 1.6$; $8.9 \cdot 10^{-18}$; selected U_k, L_k values	r461 literature/geometry audit, 6f5ea563
race values for $k = 5, \dots, 12$ and $[0.606566, 0.757783]$	r459 full-comb cleanup, 43ea44ba
raw bounds, overlaps, $0.910, 3.85 \cdot 10^{-14}, 1.89 \cdot 10^{-14}$	r460 spectral race proof, f915dd5e
(S, N) pins, $2 \cdot 10^{-12}$, census eight	r463 faithful-fold repair, 8f81005f
certified range $k = 5, \dots, 16$; $L_{16} = 2.772589 \dots$	r465 interval certificates, 74bf4ef4; r468 extension, 4a0cb4a0
low-F, ARCH, scramble, Abel-chain diagnostics	r467 overlap mechanism, a0e9aa4d
octave +0.2199; complete step -0.01180	r468 octave renormalization, 4a0cb4a0
extremal/determinant identities and strength ruling	r469 judgment, f455a22a

10 Verification anchors and exact claim boundary

The finite algebra has independent formal anchors:

- `masterCap_posSemidef_iff_Rdagger_ge_half` and `selectedWindowConeSemidef_implies_A_cap_posSemidef`;
- the compound premise `FrequentlySelectedInternalMincut`, which visibly contains `SelectedReadQuadraticRepresentation`;
- the proved conditional endpoint `internal_weil_nonneg_of_frequently_selected`;
- sealed checkers for the full-comb race, spectral reduction, narrow-band translation, end-to-end gap map, and faithful fold.

The faithful-fold pins at $k = 5, 9, 10$ have respectively $(S, N) = (19, 10), (71, 36), (79, 40)$ and reproduce the first three positive-measure Hankel entries within $2 \cdot 10^{-12}$ [C]. The formal census after exposing every known obligation is eight uses of `sorry`; this is documentation of gaps, not evidence.

Internal open steps. Round 464 proves the finite PSD implication once the exact channel representation is supplied. The two named remainders are now `SelectedReadQuadraticRepresentation` [O] and `arch_gauss_mellin_digamma_identity` [O]. The former identifies the selected matrix with the native three-channel read; the latter is the remaining classical archimedean seam.

External bridge 1: density and continuity. Extend positivity from native dyadic piecewise-linear autocorrelations to the complete admissible Weil test class [O]. This is a genuine analytic density/continuity theorem.

External bridge 2: explicit-formula identification. Identify the continued three-channel functional, with matching normalizations, with the standard Guinand–Weil explicit formula, including its spectral/zero side [O].

External bridge 3: zeta interface. Formalize the standard Weil criterion against the actual Riemann zeta function and its Riemann-hypothesis predicate [O].

Boundary. Problem 1 is the precise finite-matrix form being offered. It is open and, modulo the two internal identifications and three separately listed external bridges, has the strength of the full Weil criterion. The mechanism is not evidence; the certificates are finite statements; the residual is not a lower-strength result. The present document proves no statement about the zeros of ζ .

References

- [1] A. Weil, *Sur les formules explicites de la théorie des nombres premiers*, Meddelanden Från Lunds Univ. Mat. Sem. (1952), 252–265.
- [2] E. Bombieri, *Remarks on Weil’s quadratic functional in the theory of prime numbers, I*, Rend. Lincei Mat. Appl. **11** (2000), 183–233.
- [3] A. Borodin, *Duality of orthogonal polynomials on a finite set*, J. Statist. Phys. **109** (2002), 1109–1120.
- [4] M. Chuk, *Weil positivity in compact windows: certified two-sided bounds and a Landau–Widom decay law*, arXiv:2608.24827 (August 25, 2026), preprint.